Algebra 2 **Unit 1: Foundations — Algebra 1 Review**

Practice Test — Study Copy

Name: ____________________________ Date: ________________ Period: _________

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key. Every solve has a box of ruled space sized to the work it takes — if your work does not fill it, you probably skipped a step.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

1. Irrational number _______
2. Distributive property _______
3. Greatest common factor _______
4. Literal equation _______
5. Contradiction _______
6. Discriminant _______
7. x-intercept _______
8. Perpendicular lines _______

(A) The largest factor shared by every term, pulled out before any other factoring is attempted.

(B) The number $b^2 - 4ac$, read *before* solving to predict how many real solutions a quadratic has.

(C) A real number that cannot be written as a ratio of two integers; its decimal never terminates and never repeats.

(D) An equation true for no value at all, so its solution set is empty.

(E) Two lines whose slopes are negative reciprocals, so they meet at a right angle.

(F) The rule $a(b + c) = ab + ac$ — the only property in this unit that changes the number of terms.

(G) An equation with several letters, solved for one letter in terms of the others.

(H) A point $(a, 0)$ where a graph crosses the horizontal axis; its x-value is a zero of the function.

Part B — Multiple Choice (12 pts)

Circle the best answer.

1. Which property justifies the step $(4 + x) + 7 = 4 + (x + 7)$?
   (A) Commutative property of addition (B) Associative property of addition
   (C) Distributive property (D) Additive identity

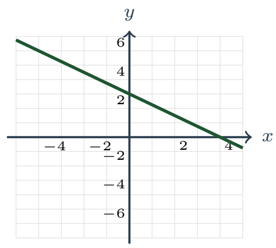
2. Simplified with no negative exponents, $\dfrac{24m^7n^{-2}}{8m^3n^2}$ is (A) $\dfrac{3m^4}{n^4}$ (B) $\dfrac{3m^4}{n^2}$ (C) $3m^4n^4$ (D) $\dfrac{16m^4}{n^4}$

3. The solution of $-3x - 5 > 7$ is (A) $x > -4$ (B) $x < -4$ (C) $x < 4$ (D) $x > 4$

4. How many *real* solutions does $x^2 - 6x + 10 = 0$ have? (A) two (B) one (C) none (D) infinitely many

5. Which relation is *not* a function?
   (A) $\{(-5, 3), (-2, 3), (1, 3), (4, 3)\}$ (B) $\{(3, -5), (3, 2), (6, 1)\}$
   (C) $\{(-2, 4), (-1, 1), (0, 0), (1, 1)\}$ (D) $y = 4x - 1$

6. **SOL-style.** Which equation represents the line shown?



(A) $y = -\frac{4}{3}x + 3$

(B) $y = -\frac{3}{4}x + 3$

(C) $y = \frac{3}{4}x + 3$

(D) $y = -\frac{3}{4}x - 3$

Part C — Short Answer & Computation (40 pts)

Show your work. A wrong answer with a right idea can still earn credit; a right answer with no work may not.

1. **Numbers and justification.**

   (a) Label each number **R** (rational) or **I** (irrational).

   $\sqrt{64}$ ______ $\sqrt{10}$ ______ $-\frac{7}{3}$ ______ $0.\overline{18}$ ______ $\sqrt[3]{5}$ ______

   (b) Name the property that justifies each step.

   $6(x-4) = 6x - 24$ ____________________________

   $9 + (x+1) = (9+x) + 1$ ____________________________

   $5x + 0 = 5x$ ____________________________

   (c) "The set of whole numbers is closed under subtraction." Disprove it.

2. **Translate and evaluate.**

   (a) Write an expression for "five more than the cube of a number n." ____________________

   (b) Evaluate $\dfrac{|a-b|}{3} + \sqrt{c}$ for $a = 4$, $b = -8$, $c = 25$.

   (c) Evaluate $\sqrt[3]{x} + 4y$ for $x = -64$, $y = \frac{3}{4}$.

3. **Exponents and polynomial operations.**

   (a) $(4a^3b^2)(-3a^4b^5)$

(b) $\dfrac{30r^5s^3}{6r^5s^7}$ (no negative exponents)

(c) $(3x-4)(2x+7)$

(d) $(5x^2-3x+1)-(2x^2+x-6)$

4. **Factor completely.** GCF first, every time.

(a) $5x^2-45$

(b) $x^2+3x-10$

(c) $3x^2+14x+8$

(d) $2x^3-14x^2+24x$

5. **Radicals and rational exponents.**

(a) $\sqrt{72} =$ ______________ (b) $\sqrt[3]{54} =$ ______________

(c) Write $\sqrt[3]{x^5}$ with a rational exponent. _______________

(d) Write $16^{3/4}$ as a radical, then evaluate it. ___________________________

(e) $\sqrt{18} + \sqrt{50} =$

6. **Linear equations and inequalities.**

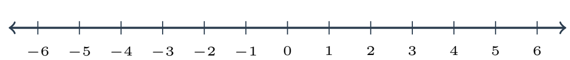
(a) Solve $4(3x - 2) = 10x + 6$.

(b) Classify each solution set as **one solution**, **no solution**, or **infinitely many**. You should not need to finish solving.

$3x + 7 = 3x - 2$ _____________________ $4(x - 3) = 4x - 12$ _____________________

(c) Solve $P = 2\ell + 2w$ for w.

(d) Solve $-3x + 2 \leq 14$, then graph the solution on the number line.



7. **Quadratic equations.**

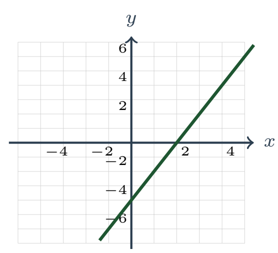
(a) Solve by factoring: $x^2 - x - 12 = 0$

(b) Solve by square roots: $(x-3)^2 = 25$

(c) Solve with the quadratic formula: $x^2 - 6x + 2 = 0$ (exact answer)

(d) *Without solving*, use the discriminant to say how many real solutions $4x^2 - 20x + 25 = 0$ has.
Discriminant: ____________________ Number of real solutions: ________________________

8. **Read the graph.** The linear function g is graphed below; each gridline is one unit.



(a) $g(-1) =$ __________ (b) Solve $g(x) = 0$: $x =$ __________

(c) That number in (b) is called the _______________ of g, and on the graph it is the point _____________.

(d) Slope: __________; y-intercept: ___________; equation of g: ___________________

(e) Is g increasing or decreasing? _______________ Domain: ______________

(f) Write the equation of the line through $(0, 5)$ **parallel** to g __________________, and the line through $(0, -1)$ **perpendicular** to g __________________.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **Three families, five questions.** Each table below is generated by a linear, a quadratic, or an exponential rule.

x	$A(x)$
0	7
1	11
2	15
3	19

x	$B(x)$
0	2
1	6
2	18
3	54

x	$C(x)$
0	3
1	4
2	7
3	12

(a) Name the family of each table and give the numerical evidence — constant difference, constant *second* difference, or constant ratio.

(b) Write a rule for each function.

$A(x) =$ ________________ $B(x) =$ ________________ $C(x) =$ ________________

(c) Which function is largest at $x = 1$? At $x = 4$? Show the values.

(d) A classmate says "B is just the biggest one." Use your answers to (c) to explain why "which is bigger" is not a fair question until someone says *where*.

(e) If B modeled a population of bacteria in hours, what domain would make sense, and why is that narrower than the domain of the rule by itself?

2. **Modeling and solution sets.** Two phone companies price a plan differently.

Northline: a one-time \$25 activation fee, then \$15 per month.

Cascade: no activation fee, \$20 per month.

(a) Write a function for the total cost at each company after m months.

$N(m) =$ ____________________ $C(m) =$ ____________________

(b) Find the number of months at which the two costs are equal. Show every step.

Name the property of equality that justifies each step above.

Step 1: ______________________________ Step 2: ______________________________

(c) Verify your answer by substituting it into *both* functions.

(d) Which company is cheaper for 3 months? For 12 months? Give the costs.

(e) Suppose Cascade changed its pricing to a \$25 activation fee plus \$15 per month. Setting the two costs equal would give $15m + 25 = 15m + 25$. What is the solution set, what is that kind of equation called, and what does it mean about the two companies?

(f) What domain makes sense for m in this situation, and why?